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RECONFIGURABLE METAMATERIAL

Abstract

A reconfigurable metamaterial formed of an assembly of building blocks including first rigid members hingedly engaged together at a first hinge, and second rigid members hingedly engaged together at a second hinge. The first rigid members are hingedly connected to respective ones of the second rigid members at third and fourth hinges. The third and fourth hinges are offset from the first and second hinges. A first elastic member is engaged to the first rigid members and biases them away from one another. A second elastic member is engaged to the second rigid members and biases them away from one another. The assembly has a deactivated configuration, wherein the third and fourth hinges are spaced apart, and an activated configuration, wherein the third hinge merges into the fourth hinge to create a contact-induced metahinge about which the first rigid members pivots relative to the second rigid members.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS [0001] The present application claims priority benefit from U.S. patent application No. 63/558,348 filed on Feb. 27, 2024, the entire contents of which are incorporated by reference herein.

TECHNICAL FIELD

[0002] The present disclosure relates generally to materials and, more particularly, to reconfigurable metamaterials.

BACKGROUND

[0003] Existing mechanical metamaterials are typically designed to either withstand loads as a stiff structure, shape morph as a floppy mechanism, or trap energy as a multistable matter, distinct behaviours that correspond to three primary classes of macroscopic solids. Their stiffness and stability are sealed permanently into their architecture, mostly remaining immutable post-fabrication due to the invariance of zero modes.

SUMMARY

[0004] In accordance with one aspect, there is provided a reconfigurable metamaterial, comprising: an assembly of building blocks secured to one another, a building block of the building blocks having: first rigid members hingedly engaged to one another at a first hinge; second rigid members hingedly engaged to one another at a second hinge, each of the first rigid members hingedly connected to a respective one of the second rigid members at a respective one of a third hinge and a fourth hinge, the third hinge and the fourth hinge being offset from the first hinge and the second hinge; a first elastic member engaged to the first rigid members and biasing the first rigid members away from one another; and a second elastic member engaged to the second rigid members and biasing the second rigid members away from one another, wherein the assembly has a deactivated configuration in which the third hinge and the fourth hinge are spaced apart from one another and has an activated configuration in which the third hinge merges into the fourth hinge to create a contact-induced metahinge about which the first rigid members pivots relative to the second rigid members.

[0005] The reconfigurable metamaterial as defined above and described herein also includes, in certain embodiments, one or more of the following features, in whole or in part, and in any combination.

[0006] In certain embodiments, the building block has a first symmetry plane intersecting both of the third hinge and the fourth hinge.

[0007] In certain embodiments, the building block has a second symmetry plane intersecting both of the first hinge and the second hinge.

[0008] In certain embodiments, in the deactivated configuration, the first rigid members and the second rigid members are spaced apart from one another by a spacing angle (2α) defined from one of the first rigid members to the other of the first rigid members around the first hinge, the first rigid members and the second rigid members having a geometry incompatibility angle (β) defined from the one of the first rigid members to a plane intersecting both of connections defined between the first rigid members and the first elastic members around one of the connections, wherein the spacing angle is smaller than two times the geometry incompatibility angle, the building block being monostable.

[0009] In certain embodiments, in the deactivated configuration, the first rigid members and the second rigid members are spaced apart from one another by a spacing angle (2α) defined from one of the first rigid members to the other of the first rigid members around the first hinge, the first rigid members and the second rigid members having a geometry incompatibility angle (β) defined from the one of the first rigid members to a plane intersecting both of connections defined between the first rigid members and the first elastic members around one of the connections, wherein the spacing angle is greater than four times the geometry incompatibility angle, the building block being bistable.

[0010] In certain embodiments, the first rigid members and the second rigid members are triangles.

[0011] In certain embodiments, the first elastic member and the second elastic member include a pair of first flexible beams and a pair of second flexible beams, the first flexible beams fixedly interconnected to one another at first proximal ends and each ending at first distal ends each connected to a respective one of the first rigid members, the second flexible beams fixedly interconnected to one another at second proximal ends and each ending at second distal ends each connected to a respective one of the second rigid members.

[0012] In certain embodiments, the building blocks are interconnected to one another to form a plurality of groups each including three building blocks, the three building blocks distributed around a central axis, the first proximal ends of the first flexible beams of each of the three building blocks of a group of the groups being fixedly connected to one another.

[0013] In certain embodiments, the groups are interconnected to one another to form a honeycomb structure.

[0014] In certain embodiments, the building blocks are interconnected to one another to form a plurality of groups each including four building blocks, the four building blocks distributed around a central axis, the first proximal ends of the first flexible beams of each of the four building blocks of a group of the groups being fixedly connected to one another.

[0015] In certain embodiments, the groups are interconnected to one another to form a matrix structure including a plurality of lines and columns.

[0016] In another aspect, there is also provided a building block for a metamaterial, comprising: first rigid members hingedly engaged to one another at a first hinge; second rigid members hingedly engaged to one another at a second hinge, each of the first rigid members hingedly connected to a respective one of the second rigid members at a respective one of a third hinge and a fourth hinge, the third hinge and the fourth hinge being offset from the first hinge and the second hinge; a first elastic member engaged to the first rigid members and biasing the first rigid members away from one another; and a second elastic member engaged to the second rigid members and biasing the second rigid members away from one another, wherein the building block has a deactivated configuration in which the third hinge and the fourth hinge are spaced apart from one another and has an activated configuration in which the third hinge merges into the fourth hinge to create a contact-induced metahinge about which the first rigid members pivots relative to the second rigid members.

[0017] The building block as defined above and described herein also includes, in certain embodiments, one or more of the following features, in whole or in part, and in any combination.

[0018] In certain embodiments, the building block has a first symmetry plane intersecting both of the third hinge and the fourth hinge.

[0019] In certain embodiments, the building block has a second symmetry plane intersecting both of the first hinge and the second hinge.

[0020] In certain embodiments, in the deactivated configuration, the first rigid members and the second rigid members are spaced apart from one another by a spacing angle (2α) defined from one of the first rigid members to the other of the first rigid members around the first hinge, the first rigid members and the second rigid members having a geometry incompatibility angle (B) defined from the one of the first rigid members to a plane intersecting both of connections defined between the first rigid members and the first elastic members around one of the connections, wherein the spacing angle is smaller than two times the geometry incompatibility angle, the building block being monostable.

[0021] In certain embodiments, in the deactivated configuration, the first rigid members and the second rigid members are spaced apart from one another by a spacing angle (2α) defined from one of the first rigid members to the other of the first rigid members around the first hinge, the first rigid members and the second rigid members having a geometry incompatibility angle (B) defined from the one of the first rigid members to a plane intersecting both of connections defined between the first rigid members and the first elastic members around one of the connections, wherein the spacing angle is greater than four times the geometry incompatibility angle, the building block being bistable.
[0022] In certain embodiments, the first rigid members and the second rigid members are triangles.
[0023] In certain embodiments, the first elastic member and the second elastic member include a pair of first flexible beams and a pair of second flexible beams, the first flexible beams fixedly interconnected to one another at first proximal ends and each ending at first distal ends each connected to a respective one of the first rigid members, the second flexible beams fixedly interconnected to one another at second proximal ends and each ending at second distal ends each connected to a respective one of the second rigid members.
[0024] In a further aspect, there is also provided a reconfigurable metamaterial, comprising: an assembly of building blocks secured to one another, a building block of the building blocks having: a first pair of first rigid members extending from first proximal ends to first distal ends, the first rigid members pivotably engaged to one another at the first proximal ends to define a first hinge; a second pair of second rigid members extending from second proximal ends to second distal ends, the second rigid members pivotably engaged to one another at the second proximal ends to define a second hinge, each of the second distal ends pivotably engaged to a respective one of the first distal ends to define a respective one of a third hinge and a fourth hinge; a first elastic member secured to both of the first rigid members, the first elastic member exerting a moment about the first hinge opposing a rotation of the first rigid members towards one another; a second elastic member secured to both of the second rigid members, the second elastic member exerting a moment about the second hinge opposing a rotation of the second rigid members towards one another; wherein the assembly has a deactivated configuration in which the third hinge and the fourth hinge are spaced apart from one another and has an activated configuration in which the third hinge merges into the fourth hinge to create a contact-induced metahinge about which the first pair of the first rigid members pivots relative to the second pair of the second rigid members.
[0025] The reconfigurable metamaterial as defined above and described herein also includes, in certain embodiments, one or more of the following features, in whole or in part, and in any combination.
[0026] In certain embodiments, the building blocks are interconnected to one another to form one or more of a matrix structure and a honeycomb structure.
[0027] The present disclosure accordingly introduces an all-in-one class of reconfigurable Kagome metamaterials that enable the in-situ reprogramming of zero modes to access the apparently conflicting properties of all classes. Through the selective activation of metahinges via self-contact, their architecture can be switched to acquire on-demand rigidity, floppiness, or global multistability, bridging the seemingly uncrossable gap between structures, mechanisms, and multistable matters. The versatile generalizations of the metahinge and reprogrammability of zero modes for a range of properties including stiffness, mechanical signal guiding, buckling modes, phonon spectra, and auxeticity, opening a plethora of opportunities for all-in-one materials and devices is showcased.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] Reference is now made to the accompanying figures in which:

[0029] FIG. **1**A illustrates primary categories of solids at the macroscopic scale classified with respect to the number of zero modes;

[0030] FIG. **1**B illustrates a nearly singular deformation mapping in a solid domain, allowing the formation of a flexible hinge (top row); topology-transformation in a multibody architecture leading to the formation of a contact-induced metahinge (middle and bottom rows).

[0031] FIG. **1**C illustrates a quarter of the architecture of FIG. **1**B annotated with key geometry parameters and its deformed configuration;

[0032] FIG. **1**D illustrates distinct energy landscapes emerging during activation for given geometric parameters; spots in distinct colors correspond to the activated state on each energy landscape; E is the Young's modulus of the elastic constituent material, and to is the out-of-plane thickness;

[0033] FIG. **1**E illustrates Contour plot of (L-L*)2/L2 in the design space defined by A′B/AA′ and OC/AA′;

[0034] FIG. **1**F illustrates a force-displacement relation describing metahinge activation; insets show deactivated (initial) and activated states; the experimental uncertainty domain is obtained by performing identical compression tests at intervals of 10 cycles of activation and deactivation;

[0035] FIG. **1**G is a schematic view of a building block for the metamaterial presented in FIG. **1**B;

[0036] FIG. **2**A illustrates a kagome metamaterial with all its metahinges being deactivated, resembling a honeycomb structure; inset at the bottom-right corner illustrates how bistable elements are joined;

[0037] FIG. **2**B illustrates the Kagome metamaterial of FIG. **2**A being fully activated; inset at the bottom-right corner is the fully activated rotation triangle;

[0038] FIG. **2**C illustrates a compression response of the metamaterial in its fully deactivated and activated state;

[0039] FIG. **2**D illustrates a lattice analogy and corresponding physical metamaterial;

[0040] FIG. **3**A illustrates selectively activated Kagome metamaterial showing a uniaxial zero mode; NO is the number of zero modes of the lattice analogy;

[0041] FIG. **3**B illustrates a linear superposition of two uniaxial zero modes and possible rotation nodes.

[0042] FIGS. **3**C to **3**E illustrate biaxial zero modes featuring symmetry and asymmetric deformation with respect to the angle bisector in white; from top to bottom, selectively activated specimens can deliver θ.sup.−e1<θ.sup.−e2, θ.sup.−e1≈θ.sup.−e2, and θ.sup.−e1>θ.sup.−e2 respectively;

[0043] FIG. **3**F illustrates mechanical XOR gate blocking the propagation of in-phase rotational signals and transmitting out-of-phase signals;

[0044] FIG. **4**A illustrates a periodic Kagome lattice with selectively added NNN bonds resembling the fully deployed state of planar multistable kirigami sheets;

[0045] FIG. **4**B illustrates an iterative optimization of energy landscape during state transition leveraging the Nudged Elastic Band method, confirming that the periodic lattice is multistable;

[0046] FIG. **4**C illustrates a configuration of the periodic lattice at the energy local maxima. d, fully collapsed state of the lattice analogy;

[0047] FIGS. **4**E to **4**G illustrate physical metastrip specimen capable of transforming its architecture into a structure (FIG. **4**E), a multistable matter (FIG. **4**F), and a mechanism (FIG. **4**G);

[0048] FIGS. **4**H to **4**J illustrate phonon spectra of periodic lattices with selectively added NNN bonds; ω* is the normalized eigenfrequency; ei axes in red indicate activated directions allowing zero modes to propagate; the bottom-left corner of each plot is the band diagram; the right column is the density plots of each eigenfrequency branch;

[0049] FIG. **5**A illustrates a square network in its fully deactivated state as a rigid structure;
[0050] FIG. **5**B illustrates a fully activated square network featuring one bulk zero mode;
[0051] FIG. **5**C illustrates a lattice analogy representing a selectively activated metamaterial;
[0052] FIGS. **5**D to **5**H illustrate snapshots of compressed metamaterial specimens in five distinct activated states; in each plot, the left and right columns display the undeformed and deformed configurations respectively; both front and side views are included; each state of the specimen is labelled with a four-digit binary number, with each digit denoting whether the corresponding row is activated (**1**) or deactivated (**0**).
[0053] FIG. **51** illustrates a force-displacement relations of selectively activated rotation-square metamaterials; dashed curves correspond to states exhibiting out-of-plane buckling, while solid curves stand for states showing in-plane buckling; friction-induced noise can be witnessed in states ‘0111’ and ‘1111’ due to the sliding between rotation squares and compression indenters;
[0054] FIG. **6**A illustrates an activation of the metahinge with C3 symmetry showing a coordination number of three; the Euler's characteristic λ transitions from −3 to −2. B;
[0055] FIG. **6**B illustrates a non-isochoric reconfiguration from a Kagome-type metamaterial to a hinged honeycomb; region shaded in yellow is the unit cell; region shaded in purple is the constituent rectangle that is kinematically determinate; insets at the bottom-right corner illustrate the topological transformation of the metahinge;
[0056] FIG. **6**C illustrates compression responses of fully deactivated and activated p6 mm metamaterial specimens;
[0057] FIG. **6**D illustrates a lattice analogy and phonon spectra for the p6 mm metamaterial; the first Brillouin Zone is shown at the top of the third column; the dashed line is the zero-frequency contour;
[0058] FIG. **6**E illustrates an activated metahinge with C4 symmetry showing a coordination number of four; the Euler's characteristic λ transitions from −4 to −3.
[0059] FIG. **6**F illustrates a non-isochoric reconfiguration from a rectangle-hinged metamaterial to a square lattice;
[0060] FIG. **6**G illustrates compression responses of corresponding fully deactivated and activated p4 mm metamaterial specimens; and
[0061] FIG. **6**H illustrates a lattice analogy and phonon spectra for the p4 mm metamaterial; the original bandgap shaded in blue vanishes upon full activation.

DETAILED DESCRIPTION

Introduction

[0062] In condensed matter physics, a zero mode (ZM) denotes a particular deformation pattern in a system incurring a fairly low energy cost. ZMs are prevalent in both natural and technological worlds across a wide spectrum of length scales, with examples encompassing but not limited to the shearing of non-viscous fluids and pentamode materials, the Guest-Hutchinson mode in the Kagome lattice, and the deployment/retraction of origami-based solar arrays. ZMs are fundamental in sealing the physical properties, e.g., stiffness and stability, of a system, as they represent the most energy-favored pathways of deformation that arise in response to an external stimulus.
[0063] As shown in FIG. **1**A, the intrinsic relation between ZMs and the fundamental physical characteristics contributes to explaining why the notion of ZMs can be used to classify a macroscopic solid into one of three classes, each with its own response that can be diametrically opposite than another. A finite number of ZMs typically defines a mechanism (left side in FIG. **1**A), which is floppy and hence apt to shape morph but often lacking load-bearing capacity. In contrast, a kinematically determinant structure has no ZMs (right side in FIG. **1**A), thus showing promising rigidity and robust stability. In between them is a multistable matter (center in FIG. **1**A), which resembles a stable structure around its energy local minima whereas behaves akin to an unstable mechanism near its energy local maxima.
[0064] Over the past two decades, each individual class of solids as shown in FIG. **1**A has inspired

the architected design of mechanical metamaterials attaining a plethora of often strikingly dissimilar mechanical characteristics. The concept of mechanism, for instance, has inspired the design of highly morphable metamaterials the notion of structure has been fundamental to boost the stiffness-to-weight ratio, and that of multistability for self-sustained reconfiguration and elastic energy trapping. Most existing metamaterials, however, lack versatility. Once fabricated with the characteristics of one sole class of solids, they cannot offer those of the others because their architecture is imprinted with an unchangeable number and pattern of ZMs. They behave as either a structure, a mechanism, or a multistable matter, hence inheriting the intrinsic stiffness and stability of the class of solids they belong to. To enable on-the-fly adjustment of stiffness or stability, reprogrammable mechanical metamaterials have been rationally designed to incorporate field-responsive constituents or reconfigurable architecture. A subset of reconfigurable metamaterials has been designed to provide adjustable kinematic determinacy that allows a transition between a structure and a mechanism, offering a remarkable degree of stiffness reprogrammability. Materials exhibiting adjustable geometry incompatibility or local confinement can also provide reprogrammability but of their stability, enabling a seamless transition between a structure and a multistable matter. Existing reprogrammable metamaterials, however, can only transition between two of the three primary classes of macroscopic solids, falling short in integrating all of them into a single unitary piece of material offering three dissimilar palettes of mechanical properties-rigidity, floppiness, and multistability.

[0065] The present disclosure introduces an all-in-one class of reconfigurable matter (which may also be referred to as reprogrammable matter) that resolves what has been so far out of reach: crossing the response boundaries between all three classes of macroscopic solids post-fabrication, and enabling the acquisition and switch on site of the traits of each class as desired. This versatility stems from a reversible process entailing the in-situ activation and deactivation of ZMs, which metamorphs the internal architecture to take on the properties of either a compliant mechanism, a rigid structure, or a multistable matter. It is demonstrated how to reprogram the ZMs to reconcile the seemingly conflicting mechanical characteristics of all three classes, hence demonstrating multifunctionality across a wide and diverse spectrum of applications. These include—but are not limited to—on-demand mechanical signal guiding, stiffness tuning, selective suppression of buckling modes, mechanical logic operations, tunable phonon spectra, and switchable auxeticity.

Activation of Metahinge Enabling Isochoric Reconfiguration

[0066] Referring to FIG. **1**B, the ZMs of a mechanical metamaterial, also known as internal mechanisms, are generally realized by connecting relatively bulky bodies via flexible hinges, hence forming an architecture where the strain energy concentrates within the flexible hinges while the relatively bulky bodies remain nearly undeformed. To activate/deactivate a ZM entails enabling the emergence or vanishing of flexible hinges. The most intuitive strategy to form a flexible hinge in a conservative system is to squeeze a finite volume of a solid domain into a tiny region Q, i.e., a flexible hinge (upper part of FIG. **1**B), a process that generally leads to a nearly singular deformation gradient F typically accompanied by excessive material stretching; this requires an abundant amount of energy typically resulting in irreversible plastic deformation or material damage.

[0067] An alternative solution to avoid excessive material stretching, while still ensuring the formation of a flexible hinge, is to introduce rationally designed perforations into the solid domain. This strategy allows material points on opposite boundaries to engage and form a flexible hinge through local rotation. The center section of FIG. **1**B shows a perforated architecture that embodies this principle; it comprises elastic beam-type components (rectangular in FIG. **1**B) and rigid components (triangular in FIG. **1**B). In the initial state, its Euler's characteristic is $\lambda=-2$. Under the squeeze of a pair of activation forces (arrows A**1** in FIG. **1**B), the architecture can undergo moderate local rotation, making the pairing edges and hinges get close and eventually merge due to self-contact. This process of activation transitions the architecture to another topological state characterized by a dissimilar Euler's characteristic, $\lambda=-1$. In this activated state, a contact-induced metahinge emerges, empowering the architecture with a rotational ZM about the metahinge. Despite the significant changes in shape and topology, there exists no excessive material stretching in the elastic beam-type components, thus realizing an energy-efficient and reversible activation process.

[0068] Referring to FIG. **1**B to **1**F, the process of metahinge activation, which features reflection symmetry, can be monostable or bistable depending on the spacing angle between pairing triangles (**2***a* in FIG. **1**B) and the geometry incompatibility (β in FIG. **1**C). Through a theoretical model, the energy landscapes during activation for varying geometric parameters, i.e., adjusting OC and A′B to alter a and β is studied.

[0069] Three main cases exist. For $\alpha<\beta$, e.g., A′B/AA′=0.7 and OC/AA′=0.6, the self-contact between pairing edges occurs prematurely and prevents the architecture from reaching its second stable state, thus resulting in a convex and monostable energy landscape (dashed line in FIG. **1**D). For $\alpha>2\beta$, e.g., A′B/AA′=0.4 and OC/AA′=2.2, the architecture is bistable, but its second stable state is a zero-energy state in the absence of self-contact (marker on the dotted curve in FIG. **1**D), meaning that a rotational ZM cannot emerge. Only for $\beta<\alpha<2\beta$, both bistability and edge contact take place, as shown by two representative energy landscapes plotted with solid curves in FIG. **1**D.

[0070] An initial version of the above architecture has been leveraged to create one-dimensional mechanical metamaterials with reprogrammable bending stiffness and local resonance. Upon activation, the architecture changes its horizontal length denoted as **2**L***-2**L (FIG. **1**B), an outcome that can result in global geometry frustration if this architecture is used as a building block of a tessellated two-dimensional metamaterial. Geometry frustration can substantially alter the volume and external shape of the material, rendering it incompatible with its original boundaries. The architecture previously reported fails to address the challenge of geometry frustration, which the present disclosure addresses. Upon activation, the horizontal span of the metahinge can be redefined to remain invariant, hence enabling an isochoric reconfiguration process that effectively eliminates undesirable geometry frustration. The expression “isochoric” refers to a process or condition in which volume remains constant. The term is used herein for a solid undergoing a specific type of transformation where the volume does not change.

[0071] By tuning OC/AA′ and A′B/AA′, it is possible to alter the length change L*-L and eventually find conditions that are length preserving. In FIG. **1**D, for instance, one of the solid lines (the one on the left) shows the energy landscape that exhibits (L*-L)/L=0.04, corresponding to a length change approximately 27% of that observed in the other energy landscape depicted by the solid line on the right, where (L*-L)/L=0.15.

[0072] Referring to FIG. **1**E, with these insights, it is possible to generate a design map where the dimensionless ratio (L-L*)2/L2 is plotted with respect to the design variables, OC/AA′ and A′B/AA′. Two curves, a=B and a=2βbound the feasible design space. Within the dark region in the feasible design space, we ultimately select a pair of values, OC/AA′=2.2 and A′B/AA′=0.7 (spot in FIG. **1**E), that guarantee nearly negligible length change (L-L*)2/L2 of approximately $1.4\times10-4$ as well as a well-merged metahinge in the activated state of the as-manufactured specimen.

[0073] FIG. **1**F illustrates the force-displacement relation of the specimen that preserved its length upon activation. A pronounced snap-through instability appears under a pair of squeezing forces, followed by an abrupt and steep rise of the reaction force Fy at the onset of self-contact (spot on the solid blue curve in FIG. **1**F). The reaction force Fy does not display a significant negative value mainly due to the viscoelasticity of the base material and the untethered loading condition, yet the activated state can be robustly preserved with self-contact. On the other hand, the metahinge is deactivated if a boundary-pulling force is applied to overcome the energy barrier and restore its original shape. To demonstrate the repeatability of the metahinge, the activation and deactivation process are cyclically performed. At intervals of every 10 cycles, the corresponding force-displacement relation for activation is recorded. This set of data is plotted as the experimental uncertainty domain shown in FIG. **1**F (zone between the dashed and solid lines). Plastic deformation may develop within the flexible hinges and stabilize after a certain number of cycles. The 101st activation exhibits a limit force of 10.7 N, a value that is approximately 81.7% that for the 1st activation. Despite this, the metahinge after cyclic usage can still promise a robust activated state, and the rotational ZM post-activation is well preserved.

[0074] Referring now to FIG. **1**G, the reconfigurable (or reprogrammable) metamaterial is built with an assembly of building blocks. One of the building blocks is shown at **10** and includes a first pair of first rigid members **11** extending from first proximal ends **11**A to first distal ends **11**B. The first rigid members are pivotably engaged to one another at the first proximal ends **11**A to define a first hinge **11**C. A second pair of second rigid members **12** extend from second proximal ends **12**A to second distal ends **12**B. The second rigid members **12** are pivotably engaged to one another at the second proximal ends **12**A to define a second hinge **12**C. Each of the second distal ends **12**B is pivotably engaged to a respective one of the first distal ends **12**B to define a respective one of a third hinge **13** and a fourth hinge **14**. The third hinge **13** and the fourth hinge **14** are offset from the first hinge **11**C and the second hinge **12**C.

[0075] A first elastic member **15** is secured to both of the first rigid members **11** and is configured to exert a moment about the first hinge **11**C opposing a rotation of the first rigid members **11** towards one another. A second elastic member **16** is secured to both of the second rigid members **12** and is configured to exert a moment about the second hinge **12**C opposing a rotation of the second rigid members **12** towards one another. In the embodiment shown, the first elastic member **15** includes two first flexible members **15**A rigidly secured to one another via a first brace **15**B. Similarly, the second elastic member **16** includes two second flexible members **16**A rigidly secured to one another via a second brace **16**B. The first and second flexible members **15**A, **16**A may be beams deformable in flexion when the first rigid members **11** and the second rigid members **12** rotate. The elastic members may be any suitable means to oppose rotation of the first and second rigid members **11**, **12** about the first and second hinges **110**, **12**C.

[0076] In the embodiment shown, the assembly including the building blocks **10** has a deactivated configuration in which the third hinge **13** and the fourth hinge **14** are spaced apart from one another and has an activated configuration in which the third hinge **13** merges into the fourth hinge **14** to create a contact-induced metahinge **16** (FIG. **1**B) about which the first pair of the first rigid members **11** pivots relative to the second pair of the second rigid members **12**.

[0077] Still referring to FIG. **1**G, the building block **10** has a first symmetry plane P**1** intersecting both of the third hinge **13** and the fourth hinge **14**. The building block **10** may have a second symmetry plane P**2** intersecting both of the first hinge **11**C and the second hinge **12**C.

[0078] In the deactivated configuration depicted in FIG. **1**G, the first rigid members and the second rigid members are spaced apart from one another by a spacing angle **2***a*. The spacing angle is defined from one of the first rigid members **11** to the other of the first rigid members **11** around the first hinge **11**C. The same angle is defined between the second rigid members **12**. The first rigid members **11** and the second rigid members **12** have a geometry incompatibility angle β, which is defined from the one of the first rigid members **11** to a plane intersecting both of connections defined between the first rigid members **11** and the first elastic members **15**A around one of the connections. This plane is parallel to the first symmetry plane P**1**. For the block **10** to be monostable, the spacing angle is smaller than two times the geometry incompatibility angle. For the block **10** to be bistable, the spacing angle is greater than four times the geometry incompatibility angle.

[0079] In some embodiments, the first rigid members **11** and the second rigid members **12** are triangles. However, other shapes may be used for these rigid members.

[0080] The first elastic member **15** and the second elastic member **16** each include a pair of first flexible beams **15**A and a pair of second flexible beams **16**A. The first flexible beams **15**A are fixedly interconnected to one another at first proximal ends and each ending at first distal ends.

Each of the distal ends is connected to a respective one of the first rigid members **11**. The second flexible beams **16**A are fixedly interconnected to one another at second proximal ends and each end at second distal ends. The second distal ends are each connected to a respective one of the second rigid members **12**.

Isochoric Reconfiguration from a Honeycomb Structure to a Kagome Mechanism

[0081] The architecture shown in FIG. **1**B can be treated as a one-dimensional bistable element that can be connected in a two-dimensional network to form a planar metamaterial capable of isochoric reconfiguration; this material is a unitary piece that can reversibly switch between a structure and a mechanism; its mechanisms of isochoric reconfiguration enables the arbitrary activation of selected metahinges without encountering geometry frustration, a distinct advantage over the existing structures.

[0082] Referring to FIG. **2**A, to study this phenomenon, a network with nodal connectivity Z=3, i.e., three bistable elements converging to each vertex is studied. Each vertex of the network is a rigid joint, disallowing free rotation. In the initial state (see fabricated finite-period specimen at the bottom of FIG. **2**A), the metamaterial resembles a hierarchical honeycomb structure with no ZMs. Upon a sequence of local activation forces, the honeycomb structure undergoes an isochoric reconfiguration where all the metahinges are activated. The activation sequence may have a negligible influence on the resulting activated state because of the isochoric characteristic of each metahinge. Consecutive activation actions are independent of each other.

[0083] In the configuration of FIG. **2**A, the building blocks **10** are interconnected to one another to form a plurality of groups **110**, each including three of the building blocks **10** described above. The three building blocks are distributed around a central axis. Namely, the first proximal ends of each of the flexible beams of the three building blocks **10** of a group of the groups are fixedly connected to one another. As shown in FIG. **2**A, the groups **110** are interconnected to one another to form a honeycomb structure.

[0084] Referring to FIG. **2**B, in the fully activated state, the honeycomb structure becomes a hinged Kagome mechanism featuring multiple floppy ZMs. In contrast to conventional multistable mechanical metamaterials, which undergo state transitions through boundary compression/tension, the disclosed metamaterial in its initial state (FIG. **2**A) is highly robust in resistance to boundary loads, as a boundary action of compression/tension is unable to induce the required counter-rotation that can activate each pair of triangles. Its state transition is triggered only when local activation forces are applied to pairing hinges that are about to merge, a characteristic implying that its initial state is to some extent protected by local geometry and is insensitive to boundary perturbations.

[0085] To characterize how the activation of multiple metahinges influences the mechanical properties of the metamaterial, compression tests are performed on specimens in both their fully deactivated (initial) and activated states. FIG. **2**C illustrates their force-displacement curves with snapshots captured during the loading process. The fully deactivated metamaterial specimen behaves as a planar structure and undergoes a homogeneous bulk deformation mode (insets i and ii of FIG. **2**C), i.e., an affine deformation response; the effective compression stiffness in this deactivated state is 2.15 N/mm. Upon being fully activated, the metamaterial specimen becomes extremely floppy; it visibly wilts under self-weight when positioned on the compression platform (inset iii of FIG. **2**C). Here the compression mainly leads to the localized deformation of the Kagome mechanism at the contact area (insets iii and iv of FIG. **2**C), delivering an effective stiffness of 0.46 N/mm. In a nutshell, from the fully deactivated state to the fully activated state, the emergence of multiple ZMs enables a remarkable reduction in compression stiffness of approximately 79%, and the deformation changes from a homogeneous bulk mode (affine) to a localized (non-affine) mode.

Lattice Analogy for Isochoric Reconfiguration of Kagome Metamaterials

[0086] To predict the number and pattern of ZMs in a Kagome metamaterial with arbitrarily activated metahinges, a lattice analogy is introduced. The fully activated Kagome metamaterial

(FIG. **2**B) is a Maxwell lattice that can be represented by an assembly of hinged triangular frameworks, as shown at the top of FIG. **2**D. Each vertex in the framework is a contact-induced metahinge allowing free (low-energy) rotation around its center. The black edges illustrated in FIG. **2**D are the nearest neighbor (NN) bonds connecting each metahinge to its nearest neighboring metahinge, showing axial stiffness kNN. If the contact-induced metahinge is deactivated, the low-energy rotation vanishes, a scenario that is equivalent to adding a pair of the next nearest neighbor (NNN) bonds, with axial stiffness kNNN, to the triangular frameworks (middle of FIG. **2**D). The lattice analogy here presented serves to capture the number and pattern of ZMs as opposed to accurately predicting the actual elasticity and inertial characteristics of the metamaterial; for this reason the value of both kNN and kNNN is now set to unity. FIG. **2**D shows the relationship between a randomly activated physical specimen and the corresponding lattice analogy, which can be leveraged to determine the number (NO) and pattern of ZMs through the calculation of the null space of the kinematic matrix.

Uniaxial and Biaxial Zero Modes Through Selective Activation of Metahinges

[0087] A fully activated Kagome metamaterial has multiple infinitesimal ZMs along its network axes, characterized by a collective counter-rotation between pairs of triangular frameworks. Due to this strong kinematic indeterminacy, multiple deformation pathways are possible in a fully activated Kagome metamaterial (FIG. **2**B). To deterministically enable one deformation pathway to emerge, we need to selectively incorporate NNN bonds so as to degenerate the Kagome metamaterial to a single-degree-of-freedom (SDOF) system, i.e. a lattice framework possessing only one ZM.

[0088] Referring to FIG. **3**A, a representative lattice analogy with only one ZM traveling along the e**1** axis, whose displacement vector is denoted as Φe**1** and depicted by the red arrows. The physical counterpart of the lattice is shown on the right of FIG. **3**A, where a rotational input on the right can be successfully delivered to the opposite end (left) through the designed ZM. We emphasize that this uniaxial ZM is infinitesimal; hence under a finite deformation, the rotation signal exhibits a decay from the source (right) to the output end (left). This type of uniaxial ZMs for mechanical signal transmission can also be realized individually along the e**2** and e**3** axes.

[0089] It is now showcased how to construct a biaxial ZM by leveraging two individual uniaxial ZMs. We first discuss the linear superposition of two ZMs along the ei and ej (i/=j) axes. In a rotation-based metamaterial, a rotation node, where the relative rotation between two adjacent bodies vanishes, can arise due to the superposition of two ZMs at their intersection boundary. A similar phenomenon can be captured in the Kagome lattice system shown in FIG. **3**B. Here NNN bonds have been strategically added to the system which has now acquired two uniaxial ZMs running through the e**1** and e**2** axes, denoted by the normalized displacement vectors Φe**1** and Φe**2** respectively. A linear combination of Φe**1** and Φe**2**, given by c**1**Φe**1**+c**2**Φe**2**, constitutes another ZM. Depending on the value of c**1**/c**2**, there exist three types of superpositioned ZMs featuring three types of rotation nodes. If c**1**/c**2**=0.5, for example, the superpositioned ZM features two rotation nodes N1; the adjacent triangles around each metahinge exhibit an identical rigid-body motion, a phenomenon implying that this metahinge is inactive. As a result, adding two pairs of NNN bonds around these two inactive metahinges has no influence on this superpositioned ZM. Adding such NNN bonds can kill the original two ZMs, Φe**1** and Φe**2**, while preserving the superpositioned ZM, 0.50e**1**+Φe**2**, a phenomenon that characterizes the unique ZM of the lattice system, i.e. the lattice framework degenerates to a SDOF system. This functionality can be leveraged to create a mechanical signal transmitter that can couple motions along two distinct axes, as described below.

[0090] The experimental results in FIGS. **3**C, **3**Dd, and **3**E attest the attainment of dissimilar modal amplitudes along two axes. For example, to demonstrate the biaxial ZM in FIG. **3**C where the rotation amplitude is larger along the e**2** axis than the e**1** axis, we first extract the four triangles outlined in a bottom part of FIG. **3**C; then we evaluate the average of their absolute rotation to

represent the modal amplitude along the e**1** axis, which is denoted as θ.sup.−e**1**. Similarly, the four triangles outlined in a top art of FIG. **3**C are used to evaluate θ.sup.−e**2**, the modal amplitude along the e**2** axis. On the right of FIG. **3**C, it is compared the experimentally obtained θ.sup.−e**1**−θ.sup.−e**2** relationship with that obtained from nonlinear Finite Element (FE) simulations of the lattice analogy. The black dashed line, with a slope of 2.0, represents the θ.sup.−e**1**-θ−e**2** relation under the assumption of infinitesimal deformation. The experimentally obtained data closely match the FE simulation results in the small-deformation regime while exhibiting a deviation as the deformation increases. This slight overestimation of θ.sup.−e**1** obtained from the FE simulation is mainly attributed to the assumed value (unity) of the axial stiffness of the NN and NNN bonds in the lattice analog model. Despite this discrepancy, the experimental results confirm that for this biaxial ZM, we can obtain θ.sup.−e**1**<θ.sup.−e**2**. On the other hand, by introducing the corresponding NNN bonds around the yellow or blue rotation nodes (left of FIGS. **3**D and **3**E respectively), we can acquire other two types of biaxial ZMs. One (FIG. **3**D) exhibits a nearly symmetric deformation with respect to the angle bisector of two axes (white dahsed line), i.e. θ.sup.−e**1**≈θ.sup.−e**2**, whereas the biaxial ZM in FIG. **3***e* has θ.sup.−e**1**>θ.sup.−e**2**.
[0091] The results obtained above can now be leveraged for mechanical logic operations. A demonstrative example is the realization of an XOR mechanical logic gate using the biaxial ZM shown in FIG. **3**D. The two independent inputs of the logic gate are the moments applied to the two rotational triangles outlined in red; their values are interpreted on the left of FIG. **3**F. The output of the logic gate is contingent on the status of the biaxial ZM. If the ZM is active, allowing the mechanical signal to be transmitted, we define the output as 1; in contrast, if the ZM is inactive, the mechanical signal is blocked, implying an output of 0. As illustrated in FIG. **3**F, only if the two input moments are out-of-phase, i.e., spinning conversely, the ZM can be activated, yielding an output of 1. Otherwise, the input moments can merely lead to an incompatible deformation localized at the area where the moments are applied. By virtue of this logical characteristic, selected metahinges can be activated in a metamaterial to act as a mechanical transmission system capable of filtering unwanted signals depending on whether the input signals are in-phase or out-of-phase.
All-In-One Reconfigerable Architecture Transforming into Either a Structure, a Multi-Stable Matter, or a Mechanism
[0092] A fully deactivated Kagome metamaterial represents a stable structure, whereas a fully activated Kagome metamaterial is an unstable mechanism. A natural question arises: can a Kagome metamaterial with a selectively activated portion of its metahinges become metastable, i.e., is it possible to reprogram the local state of each bistable metahinge to activate the global multistability of the metamaterial?
[0093] Referring now to FIG. **4**A, to address this matter, we first analyze the similarity between a Kagome lattice with selectively added NNN bonds and the fully deployed state of a planar kirigami with triangular motifs. In the lattice analogy (FIG. **4**A), the red NNN bonds and their associated triangles act akin to the Y-shaped elastic confinement observed in an existing planar kirigami; the remaining triangular sub-frameworks in the lattice analogy resemble the rotational bodies in that planar kirigami. It has been approved that the unit cell of the planar kirigami has two stable states, one deployed and one collapsed. Given the direct geometric similarity, we can anticipate that the lattice analogy presented in FIG. **4**A can also deliver an auxetic bistable transition from the deployed state to a collapsed state. To verify the bistability of this transition, we employ the Nudged Elastic Band (NEB) method and probe the minimum energy path (MEP) during state transition. If an unavoidable energy barrier exists along the MEP, the transition is bistable. Otherwise, this transition corresponds to a finite-amplitude ZM or a finite collapse mechanism. In the NEB method, we initialize the state transition path with a linear interpolation between the deployed and collapsed states, serving as our initial guess for the MEP. We iteratively update the state transition path and examine how the energy landscape evolves, as plotted in FIG. **4**B, where A

and AA stand for the initial area and area change of the unit cell respectively. We observe that the energy landscape eventually converges to the MEP represented by a red curve with a non-zero energy barrier, attesting that this state transition of the lattice analogy is bistable. FIGS. **4**C and **4**D illustrate two key states of the periodic lattice analogy: state ii, the local maxima of the MEP, and state iii, the fully collapsed state. The normalized area changes, AA/A, in states ii and iii are −0.41 and −0.75 respectively.

[0094] To experimentally demonstrate the bistable/multistable transition in a physical specimen selectively activated in the manner mentioned above, we extract a metastrip of the periodic lattice analogy, as indicated by the purple region in FIG. **4**A. A fully deactivated metastrip is a structure with no ZMs (FIG. **4**E). A selectively activated metastrip resembles the deployed state of the planar kirigami with triangular motifs, as shown on the left of FIG. **4**F. By sequentially overcoming the geometry incompatibility/energy barrier of the unit cells, we can transition the metastrip to a stable collapsed state (right of FIG. **4***f*) with an area change AA/A=−0.65; this value is in between −0.41 and −0.75 since the metastrip has exceeded the local maxima (FIG. **4**C) on the energy landscape but is prevented from reaching the fully collapsed state (FIG. **4**D) due to the presence of internal contact, or in another word, due to the intersection between NN and NNN bonds in the lattice analogy. This experimental result demonstrates that the metamaterial can undergo reversible transformations that embody its architecture with the traits of either a structure (FIG. **4**E), a multistable matter (FIG. **4**F), or a mechanism (FIG. **4**G).

Reconfigerable Phonon Spectra

[0095] Besides studying the ZMs of a selectively activated Kagome metamaterial in the static regime, we now employ the lattice analogy to demonstrate the reprogrammability of its linear phonon spectra. Illustrated at the top of FIG. **4**H-**4**J are three representative periodic lattices, each tessellated through their unit cell (highlighted in yellow) along the lattice vectors a**1** and a**2**. The unit cell has selectively added NNN bonds that allow ZMs to propagate along designated axes (red arrows). By applying Bloch's theorem and varying the wave vector within the first Brillouin zone, we can solve the eigenfrequency of this harmonic system and obtain its linear phonon spectra.

[0096] FIG. **4**H-**4**J illustrate the first five eigenfrequencies and corresponding density plots of each period lattice. The existence of zero frequencies is dependent on the wave vector [kx,ky]. For example, the lattice illustrated in FIG. **4**A, a periodic counterpart of the specimen in FIG. **3**A, has a zero-frequency contour defined by kx=0 (reciprocal space) within the lowest frequency branch ω**1**; the eigenvectors on this contour manifest localized deformation modes along the e**1** axis in direct space, and the phase velocity along this contour is zero. In total, three acoustic branches originate from [kx,ky]= [0,0]; two of them correspond to the longitudinal and shearing bulk waves, while the remaining one represents a localized wave along the e**1** axis. Once we remove relevant NNN bonds to activate another ZM along the e**3** axis (FIG. **4**I), there will be two zero-frequency contours characterized by kx=0 and kx=√3ky respectively. Now the periodic lattice manifests four acoustic branches starting at [kx,ky]= [0,0]; this lattice exhibits strong anisotropy around the [kx,ky]= [0,0] point. Upon adding a pair of NNN bonds (yellow bonds in FIG. **4**J) around the intersection node of two axes, we can eliminate the original two zero-frequency contours along kx=0 and kx=√3ky, and create a new acoustic branch manifesting a coupled deformation mode along the e**1** and e**3** axes in direct space. This emerging acoustic branch exhibits a non-zero phase velocity at [kx,ky]= [0,0], with the minimum phase velocity occurring at the angle bisector of kx=0 and kx=√3ky. Additionally, the phase velocity around the [kx,ky]= [0,0] point is less dependent on the direction of the wave vector, suggesting that the selective addition of NNN bonds can be employed to reprogram the anisotropy of the lattice in the low-frequency regime.

Reprogramming Buckling Modes in a Rotation-Square Metamaterial Via Selective Activation of Metahinges

[0097] As well known, the buckling behavior of a planar structure is governed by the competition between the in-plane stretching/compression energy and the out-of-plane bending energy. Given our metamaterial concept allows for reprogramming in-plane ZMs capable of modifying the ratio between the in-plane and out-of-plane energy, we anticipate that the selective activation of metahinges might offer a means to suppress the out-of-plane buckling of the planar metamaterial.

[0098] As a demonstrative example, we form a square network with Z=4 by connecting our bistable element (FIG. **1**B). As illustrated in FIG. **5**A, the network in its initial state is a rigid structure (N0=0). As the contact-induced metahinges are activated in the middle of each bar, the initially kinematically determinate structure turns into a rotation-square mechanism, giving rise to an auxetic bulk ZM. As shown on the right of FIG. **5**B, this bulk ZM is a finite-amplitude mechanism persisting in a large-deformation regime until the gaps between the rotation squares are fully closed. The rotation-square mechanism can be represented by a lattice analogy, where the fully activated rotation square is represented by a framework comprising four nodes and six bonds (FIG. **5**C), namely the N4B6 model. Similarly, the deactivation of the contact-induced metahinge is equivalent to adding a pair of bonds to eliminate the relative free rotation between two rotation squares. The lattice analogy becomes an SDOF mechanism only if there are no red bonds, i.e., the metamaterial is fully activated. Therefore, a partially activated metamaterial exhibits a response falling between a thick plate structure (fully deactivated) and an SDOF mechanism (fully activated), where the former buckles out-of-plane upon reaching the critical point, and the latter maintains in-plane deformation even under significant compression.

[0099] In the configuration of FIG. **5**A, the building blocks **10** are interconnected to one another to form a plurality of groups **210** each including four building blocks **10**. The four building blocks **10** are distributed around a central axis. The first proximal ends of each of the first flexible beams of the four building blocks **10** of a group of the groups being fixedly connected to one another. The groups **210** may then be interconnected to one another to form a matrix structure including a plurality of lines and columns (e.g., 2 lines by 2 columns).

[0100] FIG. **5**D-**5**H present snapshots of a compressed metamaterial specimen across five distinct activated states, and FIG. **51** shows their corresponding force-displacement relations. In the fully deactivated state '0000', the planar structure initially exhibits a monotonic force-displacement response; as the displacement reaches approximately 13.3 mm, the specimen suddenly buckles out-of-plane and undergoes a snap-back instability, a limit-point buckling phenomenon generally observed in thick plates or wide beams. In the partially activated states '0001' and '0011', the internal contact strengthens the effective bending stiffness of the local Timoshenko-type beams, a phenomenon that improves to a certain extent the global compressive stiffness of the specimen prior to the onset of instability. The out-of-plane buckling, however, still exists, and the corresponding critical displacement has rarely changed. Up to this point, the out-of-plane bending is still the energy-favored deformation mode in the post-buckling regime.

[0101] An opposite outcome to the above is obtained if the top three rows are activated (FIG. **5**G). The out-of-plane buckling is now suppressed, but the local buckling of metahinges**51** formed by self-contact is triggered at approximately 6.7 mm of compression, which leads to a discontinuous in-plane post-buckling response. Comparing FIG. **5**D-**5**G, we observe that as more rotation squares are activated, the buckling mode transitions from the global out-of-plane bending of the entire specimen to the in-plane bending of the contact-induced metahinges: the latter replaces the former as the energy-favored mode in the post-buckling regime. The critical displacement exhibits a substantial decrease as the buckling mode switches from out-of-plane to in-plane, primarily due to the characteristic dimension of the contact-induced metahinge which is much smaller than that of the entire specimen. Since the bulk ZM is not fully activated in the '0111' state, the auxetic phenomenon does not propagate across the entire specimen. As a result, a domain wall (shaded in red) emerges on the third row, delineating a boundary between the auxetic region (top) and the non-auxetic region (bottom), namely the mechanism region (top) and the structure region (bottom).

[0102] This type of domain wall has been found in rotation-square metamaterials with intentionally introduced pinning defects. In our metamaterial, the deactivation of a contact-induced metahinge is analogous to introducing a pinning defect. Our approach to introducing "defects", however, is rooted in the bistability of the local architecture, eliminating the need for manual adhesion to immobilize the defect. As a result, the versatility of our strategy enables a more convenient reprogrammability of the domain wall within a rotation-square metamaterial.

[0103] Finally, if all the metahinges have been activated, the critical displacement and force exhibit a further reduction (red curve in FIG. **51**). This is due to the complete activation of the auxetic ZM, making the entire system less stable yet with no out-of-plane buckling. The specimen now transforms into a compliant rotation-square mechanism, and its fully collapsed state is shown in the inset of FIG. **51**. In summary, this set of results shows that through the progressive activation of the rotation-square metamaterial, we can on-demand suppress the out-of-plane buckling and trigger the in-plane buckling of metahinges at a smaller scale, thus concurrently reprogramming the associated critical forces and displacements.

Generalization to Metamaterials Comprising Metahinges with a Higher Coordination Number

[0104] In the context of a fully activated metamaterial, the coordination number is defined as the average number of rotation bodies connected at each activated metahinge, e.g. the fully activated metamaterials in FIGS. **2**B and **5**H have a coordination number of two. Here, we demonstrate the creation of a metahinge featuring a higher coordination number, enabling the emergence of a larger number of ZMs upon activation. The strategy here pursued is to admit higher-order cyclic symmetry in the metahinge architecture so as to access a larger pool of metamaterial tessellations with higher coordination numbers.

[0105] The metahinge architecture illustrated in FIG. **1**B exhibits C2 cyclic symmetry yielding a corresponding set of metamaterial tessellations. If we extract its left half to create a new assembly featuring C3 symmetry, we can obtain an activated metahinge with a coordination number of three, as shown in FIG. **6**A. This architecture can then be connected to form a Kagome-type metamaterial belonging to the p6 mm group (FIG. **6**B). The activation process in this case is non-isochoric. The area of the activated unit cell shrinks to about 53% of its original value (shaded yellow regions in FIG. **6**B). The metamaterial is also able to deliver a dual response: the effective compressive stiffness, which is 4.96 N/mm in the fully deactivated state, decreases to 0.46 N/mm due to the emergence of numerous ZMs post-activation (FIG. **6**C). The fully activated specimen resembles a hinged honeycomb (inset iii of FIG. **6**C) and hence is highly flexible, leading to an initial deformation under gravity, similar to the Kagome metamaterial shown in FIG. **2**C. In this case, internal contact between pairing rotation bodies emerges in the specimen lower part due to the small spacing angle resulting from the high coordination number. This densification in the lower part results in a concentrated deformation at the upper edge of the specimen (insets iii and iv in FIG. **6**C).

[0106] The type of non-isochoric transformation described above can also be interpreted through our lattice analogy. The fully deactivated state of the metamaterial (left of FIG. **6**B) can be considered as a network hinged by a series of constituent rectangles characterized by an aspect ratio of s**1**/s**2**. A typical constituent rectangle (shaded in purple in FIG. **6**B) can be represented by the N4B6 model in the lattice analogy (left of FIG. **6**D). For non-isochoric transformations, activating the metahinges is equivalent to reducing the aspect ratio s**1**/s**2** until it approaches zero. As s**1**/s**2**.fwdarw.0, the rectangle degenerates to a bond, and the lattice analogy transforms into a hinged hexagonal lattice (FIG. **6**D). This non-isochoric transformation leads to a significant change in the phonon spectrum: the acoustic branch with the lowest group velocity degenerates to a zero-frequency contour, the bandgap (grey area) between the fourth and fifth branches is broadened, and a bandgap (blue area) between the ninth and tenth branches emerges (Supplementary Note **6**). Overall, the band structure shifts to a lower frequency regime due to the softening behavior endowed by the metahinge activation.

[0107] Similarly, we can explore higher-order of cyclic symmetry, and create for example an architecture with C4 symmetry, featuring a coordination number of four in the fully activated state, as illustrated in FIG. **6**E. The architecture can then be tessellated to build a square-type metamaterial (left of FIG. **6**F). In the fully deactivated state, the metamaterial specimen already exhibits one bulk auxetic ZM (insets i and ii FIG. **6**G), and hence it is comparably floppy under compression (blue curve in FIG. **6**G). Upon full activation, the metamaterial specimen manifests an increased number of ZMs and loses its shearing resistance. Affected by gravity and imperfections, the specimen is tilted to one side, resulting in a sequence of internal contacts between rotation bodies (insets iii and iv in FIG. **6**G); this internal contact further alters the specimen topology and substantially improves the compression stiffness (red curve in FIG. **6**G). FIG. **6**H shows the lattice analogy for this p4 mm group metamaterial. The original rectangle-hinged lattice degenerates to a square lattice upon activation. Zero frequencies emerge along the ┌-Y contour in reciprocal space, corresponding to the shearing ZMs in direct space. Similarly, the activation of metahinges can lead to the emergence and vanishing of bandgaps, as shown in FIG. **6**H. We remark that the spacing angle T in the fully activated state (FIGS. **6**A and **6***e*) is a crucial geometry parameter preventing us from reaching higher-order cyclic symmetry; τ decays exponentially with the symmetric order, and hence the internal contact between rotational bodies would disrupt the bistable activation process.

CONCLUSIONS

[0108] In summary, we have presented a class of mechanical metamaterials that feature reprogrammable zero modes enabled by the selective activation of metahinges. The architecture of the metahinge is rationally redefined over the existing literature to tackle the challenge of geometry frustration that arises during its progressive activation in a tessellated two-dimensional metamaterial. This allows the Kagome-type metamaterial to reversibly transition among a rigid structure, a compliant mechanism, and a multistable matter, hence integrating their conflicting mechanical characteristics within a single topology-transformable architecture. We also showcase the generalization of our concept to a rotation-square metamaterial and the creation of a metahinge with higher cyclic symmetry. In distinct activated states, this class of metamaterials is able to deliver stiffness reprogramming near one order of magnitude, adjustable phonon spectra, on-demand buckling mode suppression, and switchable auxeticity, providing a promising avenue for the development of all-in-one devices for application in a diverse range of engineering fields. Finally, we envision the scalability of our metahinge within a range spanning from millimeters to meters and the possibility of embedding the metahinge architecture into origami-type metamaterials, enabling on-demand activation of foldability, thus extending our concept from two-dimensional planar materials to three-dimensional bulk materials.

[0109] It is noted that various connections are set forth between elements in the preceding description and in the drawings. It is noted that these connections are general and, unless specified otherwise, may be direct or indirect and that this specification is not intended to be limiting in this respect. A coupling between two or more entities may refer to a direct connection or an indirect connection. An indirect connection may incorporate one or more intervening entities. The term “connected” or “coupled to” may therefore include both direct coupling (in which two elements that are coupled to each other contact each other) and indirect coupling (in which at least one additional element is located between the two elements).

[0110] It is further noted that various method or process steps for embodiments of the present disclosure are described in the preceding description and drawings. The description may present the method and/or process steps as a particular sequence. However, to the extent that the method or process does not rely on the particular order of steps set forth herein, the method or process should not be limited to the particular sequence of steps described. As one of ordinary skill in the art would appreciate, other sequences of steps may be possible. Therefore, the particular order of the steps set forth in the description should not be construed as a limitation.

[0111] Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. As used herein, the terms “comprises”, “comprising”, or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

[0112] While various aspects of the present disclosure have been disclosed, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible within the scope of the present disclosure. For example, the present disclosure as described herein includes several aspects and embodiments that include particular features. Although these particular features may be described individually, it is within the scope of the present disclosure that some or all of these features may be combined with any one of the aspects and remain within the scope of the present disclosure. References to “various embodiments,” “one embodiment,” “an embodiment,” “an example embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. The use of the indefinite article “a” as used herein with reference to a particular element is intended to encompass “one or more” such elements, and similarly the use of the definite article “the” in reference to a particular element is not intended to exclude the possibility that multiple of such elements may be present.

[0113] The entirety of the following disclosures are incorporated by reference herein: U.S. patent application No. 63/500,353 filed on May 5, 2023; International patent application No.: PCT/CA2022/051731 filed on Nov. 25, 2022; U.S. patent application No. 63/557,809 filed on Feb. 26, 2024; and International patent application No.: PCT/CA2024/050614 filed on May 6, 2024.

[0114] The embodiments described in this document provide non-limiting examples of possible implementations of the present technology. Upon review of the present disclosure, a person of ordinary skill in the art will recognize that changes may be made to the embodiments described herein without departing from the scope of the present technology. Yet further modifications could be implemented by a person of ordinary skill in the art in view of the present disclosure, which modifications would be within the scope of the present technology.

Claims

1. A reconfigurable metamaterial, comprising: an assembly of building blocks secured to one another, a building block of the building blocks having: first rigid members hingedly engaged to one another at a first hinge; second rigid members hingedly engaged to one another at a second hinge, each of the first rigid members hingedly connected to a respective one of the second rigid members at a respective one of a third hinge and a fourth hinge, the third hinge and the fourth hinge being offset from the first hinge and the second hinge; a first elastic member engaged to the first rigid members and biasing the first rigid members away from one another; and a second elastic member engaged to the second rigid members and biasing the second rigid members away from one another, wherein the assembly has a deactivated configuration in which the third hinge and the fourth hinge are spaced apart from one another and has an activated configuration in which the third hinge merges into the fourth hinge to create a contact-induced metahinge about which the first rigid members pivot relative to the second rigid members.

2. The reconfigurable metamaterial of claim 1, wherein the building block has a first symmetry plane intersecting both of the third hinge and the fourth hinge.

3. The reconfigurable metamaterial of claim 2, wherein the building block has a second symmetry plane intersecting both of the first hinge and the second hinge.

4. The reconfigurable metamaterial of claim 1, wherein, in the deactivated configuration, the first rigid members and the second rigid members are spaced apart from one another by a spacing angle (2α) defined from one of the first rigid members to the other of the first rigid members around the first hinge, the first rigid members and the second rigid members having a geometry incompatibility angle (β) defined from the one of the first rigid members to a plane intersecting both of connections defined between the first rigid members and the first elastic member around one of the connections, wherein the spacing angle is smaller than two times the geometry incompatibility angle, the building block being monostable.

5. The reconfigurable metamaterial of claim 1, wherein, in the deactivated configuration, the first rigid members and the second rigid members are spaced apart from one another by a spacing angle (2α) defined from one of the first rigid members to the other of the first rigid members around the first hinge, the first rigid members and the second rigid members having a geometry incompatibility angle (β) defined from the one of the first rigid members to a plane intersecting both of connections defined between the first rigid members and the first elastic members around one of the connections, wherein the spacing angle is greater than four times the geometry incompatibility angle, the building block being bistable.

6. The reconfigurable metamaterial of claim 1, wherein the first rigid members and the second rigid members are triangles.

7. The reconfigurable metamaterial of claim 1, wherein the first elastic member and the second elastic member include a pair of first flexible beams and a pair of second flexible beams, the first flexible beams fixedly interconnected to one another at first proximal ends and each ending at first distal ends each connected to a respective one of the first rigid members, the second flexible beams fixedly interconnected to one another at second proximal ends and each ending at second distal ends each connected to a respective one of the second rigid members.

8. The reconfigurable metamaterial of claim 7, wherein the building blocks are interconnected to one another to form a plurality of groups each including three building blocks, the three building blocks distributed around a central axis, the first proximal ends of the first flexible beams of each of the three building blocks of a group of the groups being fixedly connected to one another.

9. The reconfigurable metamaterial of claim 8, wherein the groups are interconnected to one another to form a honeycomb structure.

10. The reconfigurable metamaterial of claim 7, wherein the building blocks are interconnected to one another to form a plurality of groups each including four building blocks, the four building blocks distributed around a central axis, the first proximal ends of the first flexible beams of each of the four building blocks of a group of the groups being fixedly connected to one another.

11. The reconfigurable metamaterial of claim 10, wherein the groups are interconnected to one another to form a matrix structure including a plurality of lines and columns.

12. A building block for a metamaterial, comprising: first rigid members hingedly engaged to one another at a first hinge; second rigid members hingedly engaged to one another at a second hinge, each of the first rigid members hingedly connected to a respective one of the second rigid members at a respective one of a third hinge and a fourth hinge, the third hinge and the fourth hinge being offset from the first hinge and the second hinge; a first elastic member engaged to the first rigid members and biasing the first rigid members away from one another; and a second elastic member engaged to the second rigid members and biasing the second rigid members away from one another, wherein the building block has a deactivated configuration in which the third hinge and the fourth hinge are spaced apart from one another and has an activated configuration in which the third hinge merges into the fourth hinge to create a contact-induced metahinge about which the first rigid members pivot relative to the second rigid members.

13. The building block of claim 12, wherein the building block has a first symmetry plane intersecting both of the third hinge and the fourth hinge.

14. The building block of claim 13, wherein the building block has a second symmetry plane intersecting both of the first hinge and the second hinge.

15. The building block of claim 12, wherein, in the deactivated configuration, the first rigid members and the second rigid members are spaced apart from one another by a spacing angle (2α) defined from one of the first rigid members to the other of the first rigid members around the first hinge, the first rigid members and the second rigid members having a geometry incompatibility angle (β) defined from the one of the first rigid members to a plane intersecting both of connections defined between the first rigid members and the first elastic members around one of the connections, wherein the spacing angle is smaller than two times the geometry incompatibility angle, the building block being monostable.

16. The building block of claim 12, wherein, in the deactivated configuration, the first rigid members and the second rigid members are spaced apart from one another by a spacing angle (2α) defined from one of the first rigid members to the other of the first rigid members around the first hinge, the first rigid members and the second rigid members having a geometry incompatibility angle (β) defined from the one of the first rigid members to a plane intersecting both of connections defined between the first rigid members and the first elastic members around one of the connections, wherein the spacing angle is greater than four times the geometry incompatibility angle, the building block being bistable.

17. The building block of claim 12, wherein the first rigid members and the second rigid members are triangles.

18. The building block of claim 12, wherein the first elastic member and the second elastic member include a pair of first flexible beams and a pair of second flexible beams, the first flexible beams fixedly interconnected to one another at first proximal ends and each ending at first distal ends each connected to a respective one of the first rigid members, the second flexible beams fixedly interconnected to one another at second proximal ends and each ending at second distal ends each connected to a respective one of the second rigid members.

19. A reconfigurable metamaterial, comprising: an assembly of building blocks secured to one another, a building block of the building blocks having: a first pair of first rigid members extending from first proximal ends to first distal ends, the first rigid members pivotably engaged to one another at the first proximal ends to define a first hinge; a second pair of second rigid members extending from second proximal ends to second distal ends, the second rigid members pivotably engaged to one another at the second proximal ends to define a second hinge, each of the second distal ends pivotably engaged to a respective one of the first distal ends to define a respective one of a third hinge and a fourth hinge; a first elastic member secured to both of the first rigid members, the first elastic member exerting a moment about the first hinge opposing a rotation of the first rigid members towards one another; a second elastic member secured to both of the second rigid members, the second elastic member exerting a moment about the second hinge opposing a rotation of the second rigid members towards one another; and wherein the assembly has a deactivated configuration in which the third hinge and the fourth hinge are spaced apart from one another and has an activated configuration in which the third hinge merges into the fourth hinge to create a contact-induced metahinge about which the first pair of the first rigid members pivot relative to the second pair of the second rigid members.

20. The reconfigurable metamaterial of claim 19, wherein the building blocks are interconnected to one another to form one or more of a matrix structure and a honeycomb structure.